

PAPER_785: Spirograph Nebula IC 418 — UQFF Planetary Nebula Fast Wind

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Framework: UQFF (Universal Quantum Field Superconductive Framework)

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Abstract

IC 418, the “Spirograph Nebula,” is one of the most intricately structured planetary nebulae, located $\sim 2,000$ light-years away ($z \approx 0.0007$) in the constellation Lepus. Hubble WFPC2 imaging reveals a complex pattern of nested shells and radial spokes resembling a spirograph drawing. The central white dwarf ($T_{\text{eff}} \sim 36,000$ K) drives a fast stellar wind at $\sim 1,500$ km/s ($v = 1.5 \times 10^6$ m/s), ionizing the ejected AGB envelope. Under UQFF, the fast wind velocity ($v = 1.5 \times 10^6$ m/s) with standard PN B-field ($B = 10^{-5}$ T) yields $g_{\text{IC418}} \approx 1.580 \times 10^{-2}$ m/s².

1. Introduction

IC 418 represents a late-stage AGB star ($\sim 1\text{--}3 M_{\odot}$ progenitor) that ejected its outer envelope $\sim 3,000$ years ago, creating the current planetary nebula shell. The Hubble image reveals the “spirograph” interference pattern from multiple overlapping AGB pulsation shells that were ejected at slightly different angles. The central star’s fast wind at $\sim 1,500$ km/s (confirmed by UV spectroscopy) is the highest drive velocity in the slow/fast wind PN interaction model. UQFF encodes this through $v = 1.5 \times 10^6$ m/s, which is $1.5\times$ the LBV eruptive velocity (and thus $15\times$ the standard HII velocity), yielding $g = 1.580 \times 10^{-2}$ m/s².

2. Master UQFF Gravity Equation

$$g_{\text{IC418}}(r, t) = (G \times M) / r^2 \times (1 + H(z) \times t) \times (1 - E_{\text{rad}}) \times (1 + f_{\text{TRZ}}) + a_{\text{EM}}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Nebula mass (envelope)	M	$\sim 0.6 M_{\odot} = 1.193 \times 10^{30}$ kg	Hubble
Nebula radius	r	1×10^{16} m (~ 0.1 pc)	Hubble
Age	t	$\sim 3,000$ yr = 9.468×10^{10} s	Expansion age
E_rad	—	0.20	EUV photoionization loss
Redshift	z	0.0007	Distance

Parameter	Symbol	Value	Source
v_EM	v	1.5×10^6 m/s	Central star fast wind
B_EM	B	10^{-5} T	PN field
f_TRZ	—	0.05	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$g_{\text{grav}} = 6.6743\text{e-}11 \times 1.193\text{e}30 / (1\text{e}16)^2 = 7.962\text{e-}13 \text{ m/s}^2$$

Step 2: Cosmic Expansion Factor

$$H(z) \times t \approx \text{negligible} \quad (2.268\text{e-}18 \times 9.468\text{e}10 \approx 2.15\text{e-}7); \text{ factor} \approx 1.0000002$$

Step 3: Radiation Energy Loss (EUV Photoionization)

$$E_{\text{rad}} = 0.20; \quad 1 - E_{\text{rad}} = 0.80$$

Step 4: Time-Reversal Correction

$$f_{\text{TRZ}} = 0.05; \quad 1 + f_{\text{TRZ}} = 1.05$$

Step 5: Gravitational Total

$$g_{\text{grav_total}} = 7.962\text{e-}13 \times 1.0 \times 0.80 \times 1.05 = 6.689\text{e-}13 \text{ m/s}^2$$

Step 6: Aether EM Correction (Fast Wind)

$$v = 1.5 \times 10^6 \text{ m/s}, \quad B = 10^{-5} \text{ T}$$

$$a_{\text{EM}} = (1.602\text{e-}19 \times 1.5\text{e}6 \times 1\text{e-}5 / 1.673\text{e-}27) \times 11 \times 1\text{e-}12 = 1.580\text{e-}2 \text{ m/s}^2$$

Step 7: Final Solution

$$g_{\text{IC418}} = 6.689\text{e-}13 + 1.580\text{e-}2 \approx 1.580\text{e-}2 \text{ m/s}^2$$

4. Physical Interpretation

IC 418's result ($g = 1.580 \times 10^{-2}$) falls precisely between the standard HII (1.053×10^{-3}) and LBV eruptive (1.053×10^{-2}) regimes. The $1.5 \times$ velocity multiplier (1.5×10^6 vs. 1.0×10^6 m/s) directly yields a $1.5 \times$ larger result. IC 418 establishes the "fast wind PN" UQFF subcategory at $v = 1.5 \times 10^6$ m/s and will be shared with NGC 6307+7027 (PAPER_788) and M57 (PAPER_791) as the canonical planetary nebula fast-wind value.

5. Conclusions

UQFF applied to IC 418 Spirograph Nebula yields $g_{\text{IC418}} \approx 1.580 \times 10^{-2} \text{ m/s}^2$, establishing the planetary nebula fast-wind UQFF parameter class. The $1.5 \times 10^6 \text{ m/s}$ central star wind velocity is directly observed in UV spectroscopy, providing an observational anchor for the PN fast-wind UQFF subcategory.

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